Barrier-Spreading Models of Tsirelson Space

Automated Math Discovery Smoke Test

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Source and Question

This packet concerns Question 5.6 from:

> S. Garcia-Ferreira and A. C. Hernandez-Soto, *Ramsey Property and Block Oscillation Stability on Normalized Sequences in Banach Spaces*, arXiv:2009.03247.

In the notation of the source paper, if (x_i) is a normalized basic sequence and $s \in FIN^*$, then

$$\mathcal{X}(s) = \frac{\sum_{i \in s} x_i}{\left\|\sum_{i \in s} x_i\right\|}.$$

For a barrier $\mathcal{B}$, the paper defines $\mathcal{B}$-spreading models as the single-barrier case of its block asymptotic models. On page 26 the authors ask:

> Question 5.6. Is every $\mathcal{B}$-spreading model of T isomorphic to ℓ_1? where $\mathcal{B}$ is an arbitrary barrier.

This question motivates a new research line that consists of the study of the various asymptotic block models of the same normalized basic sequence. An example of this is presented in the next section where two asymptotic models will be described, one of which is spreading and the other is not.

For the next question we consider the Tsirelson space [19] denoted by T and the fact that every spreading model of T is isomorphic to ℓ_1 (for a proof we refer a reader to [14, Th. 3.5.(3)]).

Question 5.6. Is every $\mathcal{B}-$spreading model of T isomorphic to ℓ_1? where $\mathcal{B}$ is an arbitrary barrier.

The iteration applied to spreading models has been studied in the article [1]. This idea

Proof Intuition

The obstacle in Question 5.6 is that the source paper proves a $\mathcal{B}$-spreading model is a spreading sequence, but the cited theorem about Tsirelson space applies to spreading models actually generated by sequences in T. The missing bridge is small: a barrier always contains an infinite successive chain $s_1 < s_2 < \cdots$. Along such a chain the normalized block vectors $\mathcal{X}(s_n)$ form a normalized block sequence in the original space, and the defining estimate for the $\mathcal{B}$-spreading model is exactly the ordinary Brunel–Sucheston estimate for this block sequence.

A Bridge Lemma

Lemma 1. *Let $\mathcal{B}$ be a barrier on $\mathbb{N}$, let (x_i) be a normalized basic sequence in a Banach space X, and let (y_i) be a $\mathcal{B}$-spreading model of (x_i) in the sense of Garcia-Ferreira and Hernandez-Soto. Then there is a normalized block sequence (z_n) of (x_i) such that (z_n) generates (y_i) as an ordinary spreading model.*

Proof. By the barrier property, we may recursively choose an infinite chain

$$s_1 < s_2 < \cdots, \qquad s_n \in \mathcal{B}.$$

Indeed, after $s_1, \ldots, s_n$ have been chosen, apply the barrier property to the infinite tail above $\max s_n$ and take the initial segment supplied by $\mathcal{B}$. Put

$$z_n = \mathcal{X}(s_n).$$

Then (z_n) is a normalized block sequence of the basic sequence (x_i).

Let (ε_m) be a null error sequence witnessing that (y_i) is a $\mathcal{B}$-spreading model. Since $s_1 < s_2 < \cdots$, we have $\min s_n \to \infty$. Define

$$\delta_n = \sup_{m \geq n} \varepsilon_{\min s_m}.$$

Then $\delta_n \to 0$.

Fix $k \in \mathbb{N}$, indices $k \leq n_1 < \cdots < n_k$, and scalars $(a_i)_{i=1}^k \in [-1, 1]^k$. Since $s_1 < s_2 < \cdots$ and $\min s_{n_1} \geq n_1 \geq k$, the family

$$\{s_{n_1}, \ldots, s_{n_k}\}$$

belongs to $Bl^k(\mathcal{B} \restriction_{\mathbb{N}/(k-1)})$, the admissible block family appearing in the source definition. Therefore

$$\left| \left\| \sum_{i=1}^k a_i z_{n_i} \right\|_X - \left\| \sum_{i=1}^k a_i y_i \right\| \right| = \left| \left\| \sum_{i=1}^k a_i \mathcal{X}(s_{n_i}) \right\|_X - \left\| \sum_{i=1}^k a_i y_i \right\| \right| < \varepsilon_{\min s_{n_1}} \leq \delta_{n_1}.$$

This is precisely the usual spreading-model convergence condition for (z_n) with spreading model (y_i). □

Main Result

Theorem 1. *Every $\mathcal{B}$-spreading model of Tsirelson space T is isomorphic to ℓ_1, for every barrier $\mathcal{B}$.*

Proof. Let (y_i) be a $\mathcal{B}$-spreading model of T. By the lemma, (y_i) is an ordinary spreading model generated by a normalized block sequence in T. The source paper recalls the standard theorem of Odell [2, Theorem 3.5(3)] that every spreading model of Tsirelson space T is isomorphic to ℓ_1. Hence (y_i) is isomorphic to the unit vector basis of ℓ_1. □

Verification and Novelty Check

The proof uses only the source definitions, the elementary recursive selection of a successive chain inside a barrier, and Odell's theorem on spreading models of Tsirelson space. The bridge lemma is insensitive to the particular structure of T; for any Banach space X, every $\mathcal{B}$-spreading model of a normalized basic sequence in X is an ordinary spreading model of a normalized block sequence in X.

Before promotion, the run indexes were searched for the arXiv id `2009.03247`, the title, `B-spreading model`, `block asymptotic model`, and `Tsirelson`. Web searches for exact phrases around

"Is every B-spreading model of T isomorphic to ell_1"

and nearby variants found no explicit later answer. The proof should therefore be reviewed as a short direct observation using a known theorem, not as an independent reproving of the Tsirelson spreading-model theorem.

References

References

[1] S. Garcia-Ferreira and A. C. Hernandez-Soto, *Ramsey Property and Block Oscillation Stability on Normalized Sequences in Banach Spaces*, arXiv:2009.03247, 2020.

[2] E. Odell, *Stability in Banach spaces*, Extracta Mathematicae **17** no. 3 (2002), 385–425.

[3] B. S. Tsirelson, *It is impossible to imbed ℓ_p or c_0 into an arbitrary Banach space*, Funktsional. Anal. i Prilozhen. **8** (1974), 57–60.